

PETERBOROUGH TRENT UNIVERSITY, ON, Canada

WMO: 716720

Lat: 44.3511N

Lon: 78.2950W

Elev: 217

StdP: 98.75

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-19.9	-27.3	0.3	-22.4	-24.2	0.4	-18.9	7.3	-4.1	6.4	-4.4	1.2	290	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	30.8	21.3	29.0	20.6	27.5	19.6	22.8	28.2	21.9	26.9	21.1	25.5	3.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.3	25.2	20.3	15.4	24.5	19.5	14.6	23.6	68.7	28.3	65.3	26.9	62.1	25.7	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.1	5.3	4.6	-28.4	33.4	2.2	1.9	-30.0	34.7	-31.3	35.8	-32.5	36.9	-34.0	38.3		
			-28.6	24.6	2.1	1.2	-30.2	25.4	-31.4	26.1	-32.6	26.7	-34.1	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-7.3	-6.8	-1.4	5.9	13.2	17.6	20.5	19.3	15.3	8.7	2.0	-3.9
	DBStd	10.99	6.87	6.21	6.10	4.59	4.44	3.43	3.00	2.88	4.02	4.32	4.85	5.73
	HDD10.0	2272	537	470	356	137	19	1	0	0	4	78	241	431
	HDD18.3	4324	795	703	612	372	169	53	12	22	108	299	489	689
	CDD10.0	1182	0	0	2	14	117	229	327	290	163	38	2	0
	CDD18.3	193	0	0	0	0	9	31	81	53	18	1	0	0
	CDH23.3	1997	0	0	1	6	146	363	789	495	187	11	0	0
	CDH26.7	516	0	0	0	0	32	92	230	121	41	1	0	0
Wind	WSAvg	2.0	2.2	2.4	2.3	2.5	2.1	1.8	1.8	1.7	1.7	2.0	2.0	2.1
Precipitation	PrecAvg	850	72	50	52	68	83	80	83	66	77	71	74	69
	PrecMax	1147	136	96	97	128	133	186	292	130	184	176	146	124
	PrecMin	668	42	20	16	20	22	24	25	24	33	22	41	9
	PrecStd	128	23	21	24	30	28	45	54	28	32	37	28	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	10.1	20.1	24.1	30.0	31.4	32.9	32.5	30.6	25.1	17.0	11.2
		MCWB	8.1	7.8	12.2	14.7	21.0	21.1	22.5	21.5	21.8	18.6	11.6	10.0
	2%	DB	6.4	6.5	13.1	20.7	27.2	29.2	31.1	29.6	27.6	21.2	13.9	7.9
		MCWB	4.9	4.5	8.4	11.9	18.9	20.7	21.7	20.6	20.3	15.9	11.0	6.7
	5%	DB	4.1	4.1	9.9	17.5	24.6	27.2	29.3	27.7	25.5	18.5	11.8	5.6
		MCWB	2.9	2.3	6.1	9.9	16.6	19.4	20.7	19.6	19.0	14.2	9.0	4.3
	10%	DB	2.1	2.2	7.1	14.6	22.2	25.2	27.5	26.1	23.2	15.9	9.6	3.5
		MCWB	1.0	0.7	4.1	8.7	15.2	18.0	19.7	18.7	17.8	12.7	7.4	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	7.6	14.5	16.0	21.8	23.4	24.4	23.6	23.1	20.2	13.3	10.1
		MCDB	8.9	8.8	19.2	21.9	28.3	28.3	30.5	28.8	28.2	23.4	14.9	11.0
	2%	WB	5.4	4.7	9.5	13.6	19.9	22.0	23.0	22.3	21.7	17.0	11.4	6.8
		MCDB	6.1	6.2	11.7	18.3	25.7	27.7	28.5	27.0	26.2	19.4	13.7	7.7
	5%	WB	3.1	2.4	6.5	11.5	18.2	20.8	21.9	21.3	20.3	14.9	9.4	4.6
		MCDB	4.0	3.7	9.2	16.0	22.9	25.3	26.7	25.4	23.7	17.8	11.3	5.6
	10%	WB	1.1	0.8	4.1	9.4	16.4	19.5	21.0	20.2	18.7	13.1	7.6	2.4
		MCDB	2.0	2.1	6.9	13.8	20.6	23.3	25.7	24.2	22.0	15.2	9.6	3.4
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	10.5	11.4	12.9	11.9	12.3	12.1	12.4	9.9	9.1	7.6
		MCDBR	9.2	11.0	14.3	17.5	16.3	14.9	14.2	14.2	14.6	13.8	12.9	9.1
		MCWBR	8.5	9.6	9.7	9.9	8.3	7.3	6.3	6.4	7.7	9.0	9.9	8.2
	5% WB	MCDBR	9.3	10.5	13.1	14.5	14.1	12.6	12.0	11.2	12.6	12.6	11.6	8.7
		MCWBR	8.8	9.5	9.8	9.8	8.0	7.1	6.2	6.1	7.5	9.1	9.7	8.3
Clear-Sky Solar Irradiance	taub	0.300	0.302	0.316	0.352	0.373	0.409	0.409	0.399	0.362	0.349	0.307	0.280	
	taud	2.322	2.323	2.423	2.390	2.373	2.289	2.300	2.321	2.420	2.448	2.513	2.449	
	Ebn at Noon	813	886	923	911	897	861	857	853	861	824	813	813	
	Edh at Noon	80	96	100	113	118	129	127	120	100	85	67	65	
All-Sky Solar Radiation	RadAvg	1.35	2.24	3.56	4.45	5.38	5.78	5.90	5.15	4.12	2.41	1.49	1.08	
	RadStd	0.14	0.21	0.33	0.48	0.57	0.47	0.44	0.26	0.36	0.22	0.18	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page